ANUMULA ASHOK

Senior Software Engineer | Generative AI, LLM Agents & Distributed Systems
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PROFESSIONAL SUMMARY

Software engineer with 3.9 years building production systems in Java, Spring Boot, Node.js and TypeScript, with hands-on experience shipping LLM-powered products: a **LangChain-based AI agent framework** for natural-language charting and analysis with real-time SSE streaming, and **RAG/semantic search** using embeddings and Pinecone. Own a healthcare collections platform end-to-end at RagaAI (Temporal workflows, multi-tenant PostgreSQL, EHR/payment/EDI integrations) and lead a team of 3 engineers. Strong in API and microservice design, data pipelines, production debugging and code review; daily user of AI-assisted development tools (Claude Code, Cursor).

TECHNICAL SKILLS

Generative AI & Agents: LLM application development, LangChain agents, Retrieval-Augmented Generation (RAG), embeddings, vector search (Pinecone), NL-to-chart/analysis agents, streaming LLM responses (SSE), AI-assisted development (Claude Code, Cursor)
Languages & Frameworks: Java, Spring Boot, Node.js, TypeScript, JavaScript, NestJS, Next.js, React, Flutter
APIs & Distributed Systems: REST APIs, microservices, Temporal, RabbitMQ, Kafka, Redis, Hazelcast, caching, retries & idempotency, scheduled jobs, SSE
Data & Pipelines: PostgreSQL, Prisma, MongoDB (aggregations), ClickHouse, MySQL, ETL pipelines, batch processing, query tuning, indexing, multi-tenancy
Cloud & Operations: GCP, Kubernetes (GKE), CI/CD, Vercel, Grafana, Kibana, audit logging, production debugging & root-cause analysis

PROFESSIONAL EXPERIENCE

Software Development Engineer, RagaAI (Healthcare AI) *Dec 2025 – Present | Bengaluru*

- Own the **Collections platform end-to-end**: payment workflows, patient reminders, analytics and eClinicalWorks (EHR) sync, from feature design through production support.
- Integrate enterprise systems and external APIs (Stripe, Stedi EDI, EHR, RPA) through **fault-tolerant Temporal workflows** with retries, concurrency control and idempotency; the same orchestration patterns apply to agentic, long-running AI workflows.
- Built **configuration-driven, multi-tenant** services on PostgreSQL, replacing tenant-specific code paths so new client behaviour ships without code changes or redeploys.
- **Debug and resolve production issues** across microservices on Kubernetes/GKE; built internal tooling for DB migrations, audit logging and ops automation, supporting governance and traceability in a regulated domain.
- **Lead a team of 3 engineers**: break down features, guide implementation and review every change before merge; collaborate with product and platform teams.

Software Engineer, Revlitix India Pvt. Ltd. (Marketing Analytics SaaS) *Dec 2023 – Dec 2025 | Bengaluru*

- Owned the full lifecycle of a **LangChain-based AI agent framework** that turns natural-language questions into charts and analysis over customer marketing data, with **real-time SSE streaming** to the UI.
- Built **retrieval and semantic search** over platform data using embeddings and a **Pinecone** vector store to ground agent responses.
- **Refactored and migrated Java microservices to Node.js** and moved reporting to ClickHouse, delivering **~10× faster analytics** at lower infra cost.
- Designed fault-tolerant data pipelines handling **100K+ daily inserts/updates** with RabbitMQ, MongoDB and Redis caching.
- Owned backend code review as a quality gate, **raising release reliability ~20%**; led design reviews (+15% architectural compliance) and mentored junior engineers.

Associate Software Engineer, Revlitix India Pvt. Ltd. *Dec 2022 – Dec 2023 | Bengaluru*

- **Cut batch processing time ~99%** (10 minutes to seconds) through batch inserts, query optimization and indexing on hot fields.
- Built ETL pipelines integrating **12+ external platforms** (Google Ads, Meta Ads, LinkedIn, Salesforce, HubSpot, GA4, GSC), the data-ingestion foundation later used by the AI agent.
- Built a reporting engine with 15+ custom chart types and scheduled email insights that lifted platform traffic ~10%.

PROJECTS (independently built and running in production)

Chenda Vasool | chendavasool.com *Flutter · Next.js · NestJS · Prisma · PostgreSQL*

- Digital collection and account book for community organisations: cash and **UPI QR collections**, numbered PDF receipts, expenses, shared logbook, live dashboard and CSV/PDF reports.
- Shipped a **Flutter Android app (Google Play)** for field collectors and a **Next.js web app on Vercel** for treasurers, both on one NestJS/Prisma/PostgreSQL backend.
- Multi-org model with RBAC (owner, admin, collector, accountant); onboard and support live organisations.

Youth Management (People's Youth) | youthmanagment.com *Next.js · NextAuth · Prisma · PostgreSQL · Upstash Redis*

- Multi-tenant SaaS for youth associations and self-help groups to manage contributions, pooled funds and group loans, **deployed on Vercel**; used by live groups with 350+ payments and 20 loans tracked.
- Rule-driven dues and penalty engine, loan approval/repayment, PDF receipts, audit trail, Excel export, Google sign-in and RBAC.
- Automated **WhatsApp and email reminders** via scheduled cron jobs with per-group templates, timezone-aware schedules and email fallback.

EDUCATION

Master of Computer Applications (Online), Manipal University Jaipur *2025 – 2027 (in progress)*
Full Stack Web Development, Masai School | **B.Com**, Government Degree College, Hyderabad *2022 | 2017 – 2021*